

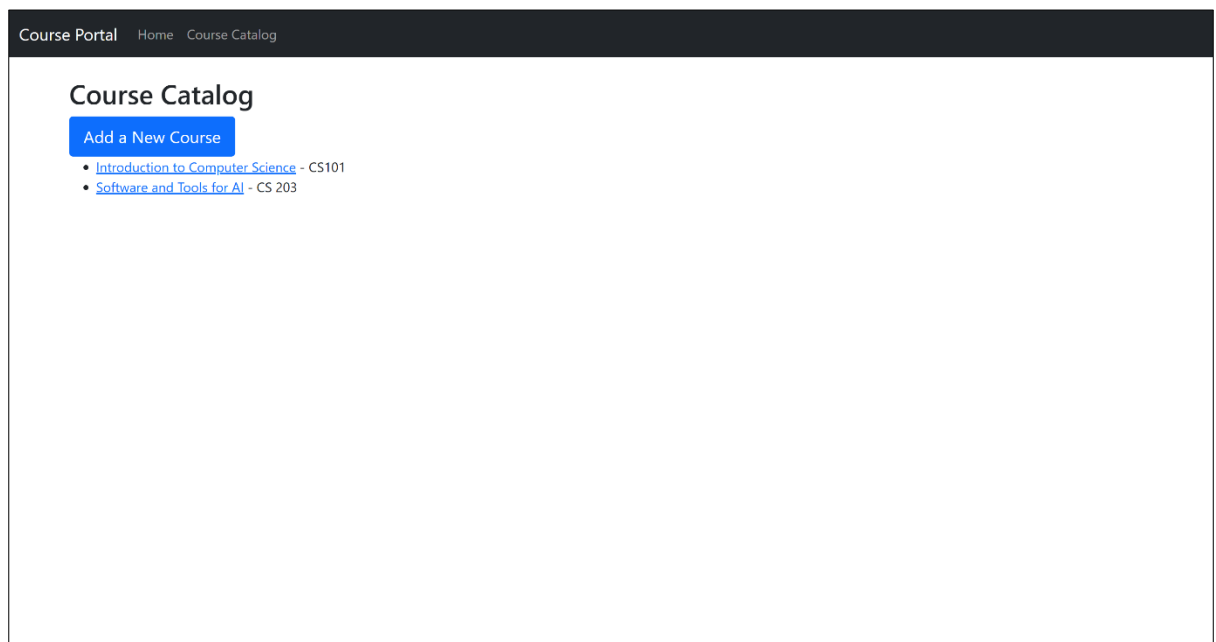
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GitHub Repo URL : [Repo URL](#)

Features Implemented:

1. Add Course to the Catalog
 - The portal now has an "Add a New Course" button_ on the catalog page. When clicked, it leads the user to a form where they can manually enter details for a new course (e.g., course name, instructor, semester).



The screenshot displays the 'Add a New Course' form within the same web application. The navigation bar and page title are consistent with the previous screenshot. The form contains several input fields for course details: 'Course Code' (filled with 'ES 335'), 'Course Name' (filled with 'Machine Learning'), 'Instructor' (filled with 'Prof. Nipun Batra'), 'Semester' (filled with 'Fall'), 'Schedule' (filled with 'K1, K2, J2'), 'Classroom' (filled with 'AB10/103'), 'Prerequisites' (filled with 'Probability, Linear Algebra'), 'Grading' (filled with 'Quizzes: 48%, Assignment'), and a 'Description' text area (filled with 'This course introduces students to the world of Machine Learning'). A final 'Add Course' button is located at the bottom of the form.

- Upon successful form submission, the course get added to the portal's course catalog, and an appropriate log message is generated.

Course Portal Home Course Catalog

Course 'Machine Learning' added successfully!

Course Catalog

[Add a New Course](#)

- [Introduction to Computer Science](#) - CS101
- [Software and Tools for AI](#) - CS 203
- [Machine Learning](#) - ES 335

```
{
  "timestamp": "2025-01-14 15:08:23,604",
  "level": "INFO",
  "message": "Saving courses to file.",
  "logger": "root",
  "trace_id": "280680128037909422437203456355877432376",
  "span_id": "2058942960500682323"
}
{
  "timestamp": "2025-01-14 15:08:23,606",
  "level": "INFO",
  "message": "{\n  'event': 'Course_added',\n  'course_data': {\n    'code': 'ES 335',\n    'name': 'Machine Learning',\n    'instructor': 'Prof. Nipun Batra',\n    'semester': 'Fall',\n    'schedule': 'K1, K2, J2',\n    'classroom': 'AB10/103',\n    'prerequisites': 'Probability, Linear Algebra, Python',\n    'grading': 'Quizzes: 48%, Assignments: 44%, Attendance: 8%',\n    'description': 'This course introduces students to the world of Machine Learning'\n  },\n  'message': '\\[Course \"Machine Learning\" added successfully.\\]'\n}",
  "logger": "root",
  "trace_id": "280680128037909422437203456355877432376",
  "span_id": "2058942960500682323"
}
```

- If required fields are missing (e.g., course name or instructor), the application log an error and notify the user with an appropriate message.

Course Portal Home Course Catalog

Missing required fields: name, instructor.

Add a New Course

Course Code:

Course Name:

Instructor:

Semester:

Schedule:

Classroom:

Prerequisites:

Grading:

Description:

```
{
  "timestamp": "2025-01-14 15:53:45,689",
  "level": "ERROR",
  "message": "{\n  'event': 'Missing_required_fields',\n  'missing_fields': ['name', 'instructor'],\n  'message': 'User attempted to submit the form with missing required fields.'\n}",
  "logger": "root",
  "trace_id": "275009390394216554749278834533785504515",
  "span_id": "9807598314230380950"
}
```

2. OpenTelemetry Tracing

- Added OpenTelemetry instrumentation to trace user requests across the following routes:
 - Course catalog page.

```
{
  "name": "course-catalog",
  "context": {
    "trace_id": "0x389a2102f04874920f5e3671dbb712ab",
    "span_id": "0x9a0540372057ccaf",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": "0xaf31fd0285703014",
  "start_time": "2025-01-14T10:29:58.210611Z",
  "end_time": "2025-01-14T10:29:58.221824Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "http.method": "GET",
    "http.url": "http://127.0.0.1:5000/catalog",
    "user.ip": "127.0.0.1",
    "courses.count": 3,
    "message": "Course catalog loaded successfully",
    "processing.duration": 0.0009996891021728516
  },
  "events": [],
  "links": [],
  "resource": {
    "attributes": {
      "telemetry.sdk.language": "python",
      "telemetry.sdk.name": "opentelemetry",
      "telemetry.sdk.version": "1.29.0",
      "service.name": "course-catalog-service"
    },
    "schema_url": ""
  }
}
```

```
{
  "name": "GET /catalog",
  "context": {
    "trace_id": "0x389a2102f04874920f5e3671dbb712ab",
    "span_id": "0xaf31fd0285703014",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": null,
  "start_time": "2025-01-14T10:29:58.208605Z",
  "end_time": "2025-01-14T10:29:58.222807Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "http.method": "GET",
    "http.server_name": "127.0.0.1",
    "http.scheme": "http",
    "net.host.name": "127.0.0.1:5000",
    "http.host": "127.0.0.1:5000",
    "net.host.port": 5000,
    "http.target": "/catalog",
    "net.peer.ip": "127.0.0.1",
    "net.peer.port": 62735,
    "http.user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36",
    "http.flavor": "1.1",
    "http.route": "/catalog",
    "http.status_code": 200
  },
  "events": [],
  "links": [],
  "resource": {
    "attributes": {
      "telemetry.sdk.language": "python",
      "telemetry.sdk.name": "opentelemetry",
      "telemetry.sdk.version": "1.29.0",
      "service.name": "course-catalog-service"
    }
  }
}
```

- Adding a new course (newly created route).

```
{
  "name": "add-course",
  "context": {
    "trace_id": "0x23263a95e44f418a0b4b356bcaebb192",
    "span_id": "0x3005842a8f94a9ae",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": "0x95a9b0bf046ccfe4",
  "start_time": "2025-01-14T10:38:16.084189Z",
  "end_time": "2025-01-14T10:38:16.089702Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "code": "ES 335",
    "name": "Machine Learning",
    "instructor": "Prof. Nipun Batra",
    "semester": "Fall",
    "schedule": "K1, K2, J2",
    "classroom": "AB10/103",
    "prerequisites": "Probability, Linear Algebra, Python",
    "grading": "Quizzes: 48%, Assignments: 44%, Attendance: 8%",
    "description": "This course introduces students to the world of Machine Learning.",
    "message": "Course 'Machine Learning' added successfully."
  },
  "events": [],
  "links": [],
  "resource": {
    "attributes": {
      "telemetry.sdk.language": "python",
      "telemetry.sdk.name": "opentelemetry",
      "telemetry.sdk.version": "1.29.0",
      "service.name": "course-catalog-service"
    }
  }
}
```

```
{
  "name": "POST /add_course",
  "context": {
    "trace_id": "0x23263a95e44f418a0b4b356bcaebb192",
    "span_id": "0x95a9b0bf046ccfe4",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": null,
  "start_time": "2025-01-14T10:38:16.083174Z",
  "end_time": "2025-01-14T10:38:16.090806Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "http.method": "POST",
    "http.server_name": "127.0.0.1",
    "http.scheme": "http",
    "net.host.name": "127.0.0.1:5000",
    "http.host": "127.0.0.1:5000",
    "net.host.port": 5000,
    "http.target": "/add_course",
    "net.peer.ip": "127.0.0.1",
    "net.peer.port": 59186,
    "http.user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36",
    "http.flavor": "1.1",
    "http.route": "/add_course",
    "http.status_code": 302
  },
  "events": [],
  "links": [],
  "resource": {
    "attributes": {
      "telemetry.sdk.language": "python",
      "telemetry.sdk.name": "opentelemetry",
      "telemetry.sdk.version": "1.29.0",
      "service.name": "course-catalog-service"
    }
  },
  "schema_url": ""
}
```

- Browsing course details.

```

{
  "name": "course-details",
  "context": {
    "trace_id": "0xc427aa1b72016884a9603cb39cf0e0b2",
    "span_id": "0x8c1d3de632d85cc2",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": "0xdbca37363998c850",
  "start_time": "2025-01-14T10:41:43.388050Z",
  "end_time": "2025-01-14T10:41:43.461723Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "course.code": "CS101",
    "course.name": "Introduction to Computer Science",
    "message": "Course details rendered successfully,"
  },
  "events": [],
  "links": [],
  "resource": {
    "attributes": {
      "telemetry.sdk.language": "python",
      "telemetry.sdk.name": "opentelemetry",
      "telemetry.sdk.version": "1.29.0",
      "service.name": "course-catalog-service"
    },
    "schema_url": ""
  }
}
{
  "name": "GET /course/<code>",
  "context": {
    "trace_id": "0xc427aa1b72016884a9603cb39cf0e0b2",
    "span_id": "0xdbca37363998c850",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": null,
  "start_time": "2025-01-14T10:41:43.387052Z",
  "end_time": "2025-01-14T10:41:43.461723Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "http.method": "GET",
    "http.server_name": "127.0.0.1",
    "http.scheme": "http",
    "net.host.name": "127.0.0.1:5000",
    "http.host": "127.0.0.1:5000",
    "net.host.port": 5000,
    "http.target": "/course/CS101",
    "net.peer.ip": "127.0.0.1",
    "net.peer.port": 49882,
    "http.user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36",
    "http.flavor": "1.1",
    "http.route": "/course/<code>",
    "http.status_code": 200
  },
  "events": [],
  "links": [],
  "resource": {
    "attributes": {
      "telemetry.sdk.language": "python",
      "telemetry.sdk.name": "opentelemetry",
      "telemetry.sdk.version": "1.29.0",
      "service.name": "course-catalog-service"
    },
    "schema_url": ""
  }
}

```

- Created meaningful spans for key operations (e.g., rendering the course catalog, handling form submissions, etc.).

```

131 @app.route('/catalog')
132 def course_catalog():
133     start_time = time.time()
134     with tracer.start_as_current_span("course-catalog", kind=SpanKind.SERVER) as span:
135         route_request_counter.add(1, {"route": "/catalog", "method": request.method})
136
137         courses = load_courses()
138         span.set_attributes({
139             "http.method": request.method,
140             "http.url": request.url,
141             "user.ip": request.remote_addr,
142             "courses.count": len(courses),
143             "message": "Course catalog loaded successfully."
144         })
145
146         processing_duration = time.time() - start_time
147         span.set_attribute("processing.duration", processing_duration)
148

```

```

{
  "name": "course-catalog",
  "context": {
    "trace_id": "0x389a2102f04874920f5e3671dbb712ab",
    "span_id": "0x9a0540372057ccaf",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": "0xaf31fd0285703014",
  "start_time": "2025-01-14T10:29:58.210611Z",
  "end_time": "2025-01-14T10:29:58.221824Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "http.method": "GET",

```

```

159 @app.route('/add_course', methods=['GET', 'POST'])
160 def add_course():
161     with tracer.start_as_current_span("add-course", kind=SpanKind.SERVER) as span:
162         route_request_counter.add(1, {"route": "/add_course", "method": request.method})
163
164         if request.method == 'POST':
165             all_fields = ['code', 'name', 'instructor', 'semester', 'schedule', 'classroom', 'prerequisites', 'grading', 'descripti
166             required_fields = ['code', 'name', 'instructor']
167             course_data = {field: request.form.get(field) for field in all_fields}
168             missing_fields = [field for field, value in course_data.items() if not value and field in required_fields]
169

```

```

{
  "name": "add-course",
  "context": {
    "trace_id": "0x23263a95e44f418a0b4b356bcaebb192",
    "span_id": "0x3005842a8f94a9ae",
    "trace_state": "[]"
  },
  "kind": "SpanKind.SERVER",
  "parent_id": "0x95a9b0bf046ccfe4",
  "start_time": "2025-01-14T10:38:16.084189Z",
  "end_time": "2025-01-14T10:38:16.089702Z",
  "status": {
    "status_code": "UNSET"
  },
  "attributes": {
    "code": "ES 335",
    "name": "Machine Learning",

```

- Added trace attributes (e.g., user IP, request methods, metadata about courses).

```

"attributes": {
  "http.method": "GET",
  "http.server_name": "127.0.0.1",
  "http.scheme": "http",
  "net.host.name": "127.0.0.1:5000",
  "http.host": "127.0.0.1:5000",
  "net.host.port": 5000,
  "http.target": "/catalog",
  "net.peer.ip": "127.0.0.1",
  "net.peer.port": 62735,
  "http.user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/131.0.0.0 Safari/537.36",
  "http.flavor": "1.1",
  "http.route": "/catalog",
  "http.status_code": 200
},

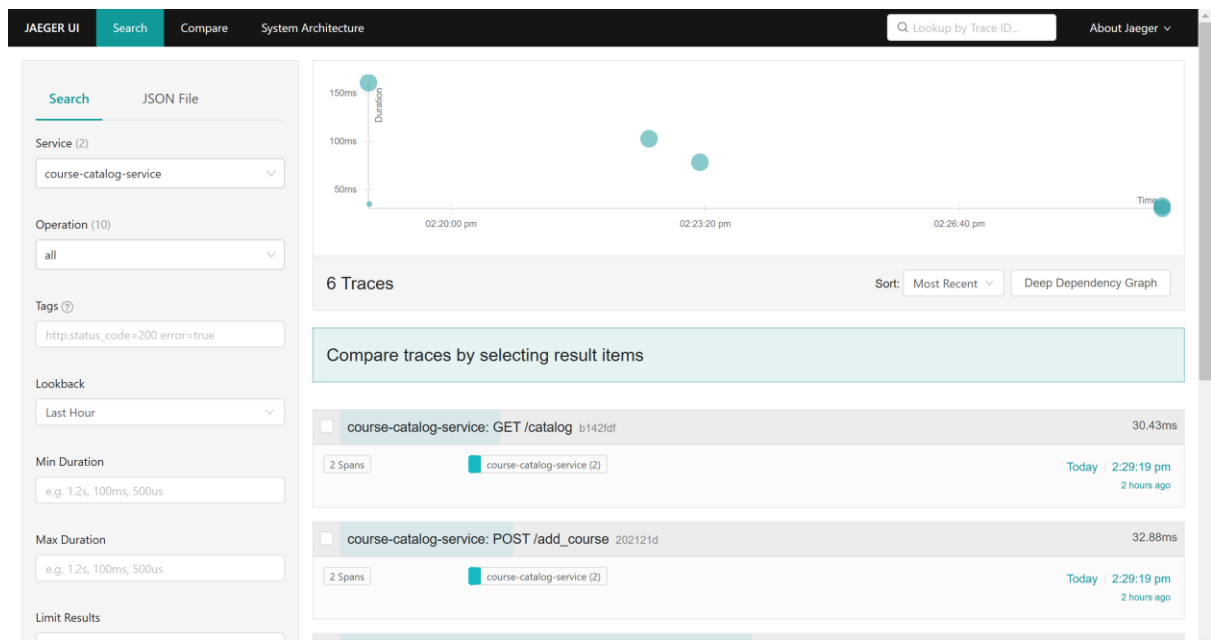
```

```
},
"attributes": {
  "code": "ES 335",
  "name": "Machine Learning",
  "instructor": "Prof. Nipun Batra",
  "semester": "Fall",
  "schedule": "K1, K2, J2",
  "classroom": "AB10/103",
  "prerequisites": "Probability, Linear Algebra, Python",
  "grading": "Quizzes: 48%, Assignments: 44%, Attendance: 8%",
  "description": "This course introduces students to the world of Machine Learning",
  "message": "Course 'Machine Learning' added successfully."
},
}
```

3. Exporting Telemetry Data to Jaeger

- Set up Jaeger as the tracing backend and ensured that telemetry data is exported to Jaeger.

```
53 # Jaeger Exporter Setup
54 jaeger_exporter = JaegerExporter(
55     agent_host_name="localhost",
56     agent_port=6831,
57 )
58 span_processor = BatchSpanProcessor(jaeger_exporter)
59 trace.get_tracer_provider().add_span_processor(span_processor)
```



```

"metrics": [
  {
    "name": "route_requests_total",
    "description": "Counts the number of requests for each route",
    "unit": "1",
    "data": {
      "data_points": [
        {
          "attributes": {
            "route": "/add_course",
            "method": "GET"
          },
          "start_time_unix_nano": 1736854423944554200,
          "time_unix_nano": 1736854469603908400,
          "value": 1,
          "exemplars": [
            {
              "filtered_attributes": {},
              "value": 1,
              "time_unix_nano": 1736854423944554200,
              "span_id": 18230787083351510496,
              "trace_id": 280156953920057982612437115496848351480
            }
          ]
        }
      ],
      {
        "attributes": {
          "route": "/add_course",
          "method": "POST"
        },
        "start_time_unix_nano": 1736854426087259300,
        "time_unix_nano": 1736854469603908400,
        "value": 1,
        "exemplars": [
          {
            "filtered_attributes": {},
            "value": 1,
            "time_unix_nano": 1736854426087259300,
            "span_id": 457064285008934997,
            "trace_id": 337551533995105887585048168578579291345
          }
        ]
      }
    ]
  }
]

```

```

    },
    {
      "name": "error_count_total",
      "description": "Counts the number of different errors",
      "unit": "1",
      "data": {
        "data_points": [
          {
            "attributes": {
              "error_type": "course_not_found"
            },
            "start_time_unix_nano": 1736855060913736800,
            "time_unix_nano": 1736855079602038100,
            "value": 1,
            "exemplars": [
              {
                "filtered_attributes": {},
                "value": 1,
                "time_unix_nano": 1736855060913694600,
                "span_id": 10609256885517304265,
                "trace_id": 308134535795925271166632943628159096392
              }
            ]
          }
        ],
        "aggregation_temporality": 2,
        "is_monotonic": true
      }
    }
  ]
}

```



```
    "trace_id": 0, "span_id": 0 }
  {
    "name": "index-route",
    "context": {
      "trace_id": "0x808c00a1f8e84de9d50db85ee148fe6c",
      "span_id": "0xe7a32008ab583dea",
      "trace_state": "[]"
    },
    "kind": "SpanKind.SERVER",
    "parent_id": "0xfa45cf6175d7c3a4",
    "start_time": "2025-01-14T11:47:27.484509Z",
    "end_time": "2025-01-14T11:47:27.485348Z",
    "status": {
      "status_code": "UNSET"
    },
    "attributes": {
      "http.method": "GET",
      "http.route": "/",
      "user.ip": "127.0.0.1",
      "message": "Index page rendered successfully.",
      "processing_time": 0.0002486705780029297
    },
    "events": [
      {
        "name": "Rendering index page",
        "timestamp": "2025-01-14T11:47:27.484544Z",
        "attributes": {}
      }
    ]
  },
]
```